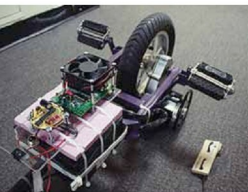


Self-balancing electric unicycle

MIT student Stephan Boyer has created a self-balancing electric unicycle. Called Bullet, the cycle packs a custom MIG-welded steel chassis, two 7Ah 12V batteries, an ATmega328P micro-controller chip and a 450-watt electric motor. It leans forward to accelerate and leans back to brake. The top speed is about 24 kilometres per hour, and the cycle easily travels eight kilometres on a single charge.



Bullet goes 8 km on a single charge

Bullet integrates readings from the gyro and accelerometer using a complementary filter. To balance, the angle estimate is fed through a proportional-integral-derivative loop (with no integral term). The loop runs at 625 Hz. The output from this stage determines the duty cycle of a 1.22kHz pulse-width modulation signal, which is connected to the H-bridge. The code is written in 'C'.

Charge your cellphone with water

Swedish company MyFC has developed a power source that converts hydrogen into electricity to charge mobile devices.

PowerTrek uses plain tap water to charge mobile devices. It is pocket-size, lightweight and fuel-cell-driven. Made from foils and adhesives, FuelCell-Stickers form a flexible assembly less than 2.75mm thick. Since the hydrogen



PowerTrek uses plain tap water to charge mobile devices

fuel can be supplied from several alternative sources, the system is 'flexifuel.'

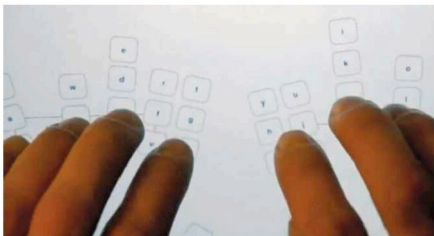
The fuel cell inside PowerTrek is a completely passive system. Without fans or pumps, it silently converts hydrogen into electricity via its proton exchange membrane. The chemistry process is safe, controllable and eco-friendly, and the only byproduct from the fuel cell is a little water vapour. To operate, hydrogen must be supplied to the fuel cell, and the fuel cell exposed to the open air.

Liquid keyboard for touchscreen tablets

Typing on a tablet's touchscreen was never this easy. Thanks to 'LiquidKeyboard' technology developed

by computer systems researcher Christian Sax along with his colleague Hannes Lau at the University of Technology in Sydney, a keyboard can morph around the user's fingers to type on the tablet's touchscreen using both hands.

The technology follows the traditional QWERTY keyboard system. A virtual keyboard is developed with the LiquidKeyboard software, which adapts automatically to the user's hand physiology, such as hand size and finger position. As soon as the first four fingers touch the surface—in one fluid motion—an entire keyboard is constructed. The system senses the pressure and position of the user's fingers on the touchscreen.



Liquid keyboard for tablets